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power output, which offers significant advantages for UAV aircraft. Additionally, the engine rotors cannot seize, since rotor casings expand more than the rotors; the engine is not susceptible to shock-cooling during descent of the UAV.

Electric and fuel cells. Solar-electric UAVs hold potential for unlimited flight and many of the small UAVs are equipped with these propulsion systems. Electric UAVs kept aloft indefinitely by laser power beaming technology represent another proposed propulsion solution for endurance.

The electric fuel cell propulsion system for UAV applications has low noise and signature, while taking advantage of hydrogen, a high-energy fuel. Fuel cells create an electrical current when they convert hydrogen and oxygen into water and heat. The fuel cell has about four times the efficiency of a comparable internal combustion engine and the system provides seven times the energy in the equivalent weight of batteries.

Future applications

The UAV market, whether for commercial, industrial or military customers, is growing rapidly and is expected to expand for several years to come. These are examples of the development of maternet where computer-powered drones carry objects much like Internet carries data.

In future, drones will power cell phones and television services. Functioning as pseudo-satellites, UAVs could be deployed to a specific location and 'stay there for days.' The benefits of these satellites are they are less expen-

sive and less susceptible to losing power compared to conventional satellites. Hydrogen-fuelled drones that can stay aloft for up to four or more days at a time, are currently being tested as a platform for telecommunications payloads. It is expected that telecom-equipped drones could revolutionise the

wireless market.

Control and guidance of UAVs

The control and guidance of these small unmanned aircraft are certainly impressive. Wireless miniature sensors are being used extensively. A variety of electronic subsystems have been specially developed for UAVs such as the radios that communicate video and data signals back to a base camp.

An operator on the ground, with only a laptop computer and tactical radio, could command what was described as a 'swarm' of UAVs without using a ground-control station.

Artificial intelligence is advancing to the point where the aircraft are easily capable of taking off, landing and flying themselves. Then they simply have to be instructed as to their mission.

Common-airborne sense and avoid technology is being incorporated into UAVs that are used for intelligence, surveillance and reconnaissance missions.

Conclusion

Taking advantage of improvements in materials, structures, batteries and electric motors, as well as microprocessor advances, many companies are coming up with innovative configurations of UAVs.

The knowledge and skill levels of system integration of multidisciplinary technologies are available indigenously. It is time for the Indian industry to take advantage of these skills. ●

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